Ingeteam

THREE-PHASE INVERTER MANAGING USING MODBUS-RTU

AAS1000ICB08_

Ingeteam Power Techonology S.A.

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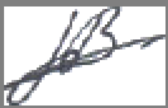

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1. INTRODUCTION

This report contains some documentation about all Ingecon Sun Three-phase family. The following information explains on one hand how to get information of the inverter and on the other hand how to manage the inverter using the MODBUS RTU protocol.

2. INPUT REGISTERS

Relevant data in each 3-phase Ingecon Sun is identified as a 16 bit **Input Registers** in the 30001 Modbus Range.

Dir. H.R.	Description	Signed	Since firmware versión
30001	Total Energy (High)		_
30002	Total Energy (Low) **30001-30002:** 30001 and 30002 registers are the High and Low parts, respectively, of the 32 bit variable that stores, in **kWh**, the total Value of produced energy over its lifecycle.		_
30003	Hours up and running (High)		_
30004	Hours up and running (Low) **30003-30004:** 30003 y 30004 registers are the High and Low parts, respectively, of the 32 bit variable that stores the number of hours that the inverter has been running.		_
30005	Number of connections to grid (High)		_
30006	Number of connections to grid (Low) **30005-30006:** 30005 y 30006 registers are the High and Low parts, respectively, of the 32 bit variable that stores the number of times that the inverter has connected to the grid since first startup.		_
30007	Inverter alarms (high)		_
30008	Inverter alarms (low) **30007-30008** Inverter alarm codes. -. ALARMA_FRED (0x0001): Grid frequency outside limits. -. ALARMA_VRED (0x0002): Grid voltage outside limits. -. ALARMA_PI_ANA (0x0004): Current PI saturation. -. ALARMA_RESET_WD (0x0008): inverter reset by Watch Dog -. ALARMA_IRED_EFICAZ (0x0010): Excessive RMS grid current -. ALARMA_TEMPERATURA (0x0020): Max. Temperature reached. -. ALARMA_LEC_ADC (0x0040): A/D converters reading error -. ALARMA_IRED_INSTA (0x0080): AC overcurrent -. ALARMA_PROT_AC (0x0100): AC protections -. ALARMA_PROT_DC (0x0200): DC protections -. ALARMA_PARO_AISL_DC (0x0400): DC Isolation Failure. -. ALARMA_FRAMA (0x0800): Failure in power electronics -. ALARMA_PARO_MANUAL (0x1000): Manual stop -. ALARMA_CONFIG (0x2000): Change in configuration -. ALARMA_VIN (0x4000): Excessive input voltage -. ALARMA_VPV_MED_MIN (0x8000): Tensión mínima en la entrada		_
30009	Vdc **:** DC input voltaje, in **Volt**s		_

30010	Idc **:** Average input current estimation. In **Amp**s	yes	_
30011	Vac1**:** Grid RMS Voltage, phase 1, in **Volt**s		_
30012	Vac2**:** Grid RMS Voltage, phase 2, in **Volt**s		_
30013	Vac3**:** Grid RMS Voltage, phase 3, in **Volt**s		_
30014	Iac1**:** Grid RMS Current, phase ,1 In **Amp**s		_
30015	Iac2**:** Grid RMS Current, phase 2, In **Amp**s		_
30016	Iac3**:** Grid RMS Current, phase 3, In **Amp**s		_
30017	Cos(φ) **:** in thousandths (φ, delay angle between AC voltage and current)		_
30018	Sign of sin(φ). 0 and 1 are for positive and negative values, respectively.		_
30019	Power **:** Alternating output power generated by the INGECON-SUN, in **tens** of **Watt**.	yes	_
30020	Grid frequency in **hundredths** of **Hz**		_
30021	Current year.		_
30022	Current month		_
30023	Current day		_
30024	Current hour		_
30025	Current minute		_
30026	Current sec		_
30027	Sun position (degrees x10)		_
30028	Suntracker 1, Mode		_
30029	Suntracker 1, motor state		_
30030	Suntracker 1, motor position (degrees x 10)		_
30031	Suntracker 1, motor position order (degrees x10)		_
30032	Suntracker 1, Alarms		_
30033	Suntracker 1, automatic tracking offset (degrees x 10)		_
30034	Suntracker 1, automatic step (degrees x 10)		_
30035	Suntracker 2, Mode		_
30036	Suntracker 2, motor state		_
30037	Suntracker 2, motor position order (degrees x 10)		_
30038	Suntracker 2, motor position order (degrees x 10)		_
30039	Suntracker 2, Alarms		_
30040	Suntracker 2, automatic tracking offset (degrees x 10)		_
30041	Suntracker 2, automatic step (degrees x 10)		_
30042	Analog Input 1		_
30043	Analog Input 2		_
30044	Analog Input 3		_
30045	Analog Input 4		_
30046	Analog Input 5 – PT100 nº 1		_
30047	Analog Input 6 – PT100 nº 2 Analog Inputs 1 to 6 contain values between 0 and 4095 (12 bit AD converters), samples provided if optional Analog Inputs card (ref. AAP0016) is installed. Read 0 if not installed. 1 to 4 are general purpose inputs, 5 and 6 are designed for PT100		_

	temperature probes.		
30048 to 30058	Reserved		_
30059	Energy Resetable Counter (Hi).		_O
30060	Energy Resetable Counter (low) **3003B-3003C** registers represent the High and Low parts, respectively, of the 32 bit resettable variable that stores, in kWh, the value of produced energy since last reset.		_O
30061	Hours up and running (high)		_O
30062	Hours up and running (low) **3003D-3003E** registers are the High and Low parts, respectively, of the 32 bit resettable variable that stores the number of hours that the inverter has been running since last reset.		_O
30063	Number of connections to grid (high)		_O
30064	Number of connections to grid (low) **3003F-30040** registers are the High and Low parts, respectively, of the 32 bit resettable variable that stores the number of times that the inverter has connected to the grid since last reset.		_O
30065	Instantaneous inverter alarms (high)		_O
30066	Instantaneous inverter alarms (low) (this variable is reset periodically every second) **30041-30042:** Inverter alarms code. -. ALARMA_FRED (0x0001): Grid frequency out of range. -. ALARMA_VRED (0x0002): Grid voltage out of range. -. ALARMA_PI_ANA (0x0004): Current PI Saturation. -. ALARMA_RESET_WD (0x0008): Inverter reset by Watch Dog -. ALARMA_IRED_EFICAZ (0x0010): Grid current effective excessive. -. ALARMA_TEMPERATURA (0x0020): Electronic temperature. -. ALARMA_LEC_ADC (0x0040): Alarm in Reading the ADC -. ALARMA_IRED_INSTA (0x0080):Instantaneous overcurrent in AC -. ALARMA_PROT_AC (0x0100): AC protections. -. ALARMA_PROT_DC (0x0200): DC protections. -. ALARMA_PARO_AISL_DC (0x0400): Isolation DC. -. ALARMA_FRAMA (0x0800): Failure in the power electronic -. ALARMA_PARO_MANUAL (0x1000): Manual stop -. ALARMA_CONFIG (0x2000): Change in the configuration -. ALARMA_VIN (0x4000):Input voltage -. ALARMA_VPV_MED_MIN (0x8000): Minimum voltage in the input		_O
30067	Inverter alarms mainteined (high)		_O
30068	Inverter alarms mainteined (low) (This variable is reset whenever the inverter		_O

	connects to grid) **3000E-3000F:** CInverter alarm codes are the same than in the parameter instantaneous inverter alarms.		
30069	QAC (Reactive power DIV 10)	yes	_O
30070	Zpos (Solar field impedance, POS-EARTH)		_O
30071	Zneg (Solar field impedance, NEG-EARTH)		_O
30072	Power electronics temperature	Yes	_O
30073	Control electronics temperature	Yes	_O
30074	Inverter state: **30074:** Inverter state. This variable reflects the inverter state and the 8 bits with lower weight means the following: 8 lower weight bit: - 0: Initial state - 1: Initial magnetization state - 2: State grid connected - 3: State error. 8 Bit more weight: - Bit 0: 1:Stop / 0:Run - Bit 1: Blocked - Bit 2: Grid Fault detected.		_O
30075	VpvN: Voltage NEGATIVE- EARTH		_O
30076	VpvP: Voltage POSITIVE - EARTH		_O
30077	Nominal power DIV10		_O
30078	Pos. Last Stop reason (0 to 4) This variable indicates which of the last 5 stop reason is the newest.		_O
30079	Year stop reason 0		_O
30080	Month stop reason 0		_O
30081	Day stop reason 0		_O
30082	Hour stop reason 0		_O
30083	Minute stop reason 0		_O
30084	Stop reason 0 -1 MOTIVO_PARO_VIN: Input voltage very high -2 MOTIVO_PARO_FRED: Grid frequency out of range -3 MOTIVO_PARO_VRED: Grid voltage out of range -4 MOTIVO_PARO_VARISTORES:Failure in protection varistors -5 MOTIVO_PARO_AISL_DC: Isolation failure in solar field -6 MOTIVO_PARO_IAC_EFICAZ: RMS output curren higher than the limit -7 MOTIVO_PARO_TEMPERATURA: Stop because of high temperature -8 MOTIVO_PARO_LATENCIA_SPI: Communication error in the SPI bus -9 MOTIVO_PARO_CONFIGURACION: Stop because of configuration change -10 MOTIVO_PARO_PARO_MANUAL: Manual stop inverter -11 MOTIVO_PARO_BAJA_VPV_MED: Stop due to low voltage in the solar field -12 MOTIVO_PARO_HW_DESCX2: Hardware error (NOT IN USE) -13 MOTIVO_PARO_FRAMA3: Failure in branch 3		_O

	-14 MOTIVO_PARO_MAX_IAC_INST: Instantaneous output current higher than the limit. -15 MOTIVO_PARO_CARGA_FIRMWARE: Stop by firmware load -16 MOTIVO_PARO_REDUNDANTE: Error from the redundant DSP -17 MOTIVO_PARO_PROTECCION_PIB: Error in the PIB protection (Only for Italy) -18 MOTIVO_PARO_ERROR_LEC_ADC: Intern error in the ADC -19 MOTIVO_PARO_CONSUMO_POTENCIA: Stop due to power consumption -20 MOTIVO_PARO_FUS_DC: DC fuses melt -21 MOTIVO_PARO_TEMP_AUX: Error in the temperature auxiliary protection -22 MOTIVO_PARO_PROT_AC: Failure in the AC controller -23 MOTIVO_PARO_MAGNETO: Trigger from a thermomagnetic protection -24 MOTIVO_PARO_CONTACTOR: Error in the grid connection contactor. -25 MOTIVO_PARO_RESET_WD: Reset WD from DSP. -26 MOTIVO_PARO_PI_ANA_SAT: Saturation in the current control -27 MOTIVO_PARO_LATENCIA_ADC: Latent error in the ADC -28 MOTIVO_PARO_ERROR_FATAL: Fatal error from the power electronics -29 MOTIVO_PARO_FRAMA1: Failure in branch 1 -30 MOTIVO_PARO_FRAMA2: Failure in branch 2		
30085	Year Stop reason 1		_O
30086	Month Stop reason 1		_O
30087	Day Stop reason 1		_O
30088	Hour Stop reason 1		_O
30089	Minute Stop reason 1		_O
30090	Stop reason 1 (Defined before)		_O
30091	Year stop reason 2		_O
30092	Month stop reason 2		_O
30093	Day stop reason 2		_O
30094	Hour stop reason 2		_O
30095	Minute Stop reason 2		_O
30096	Stop reason 2 (Defined before)		_O
30097	Year stop reason 3		_O
30098	Month stop reason 3		_O
30099	Day stop reason 3		_O
30100	Hour stop reason 3		_O
30101	Minute stop reason 3		_O
30102	Stop reason 3 (Defined before)		_O
30103	Year stop reason 4		_O
30104	Month stop reason 4		_O
30105	Day stop reason 4		_O
30106	Hour stop reason 4		_O
30107	Minute stop reason 4		_O
30108	Stop reason 4 (Defined before)		_O
30109	Remaining time to connect (seconds)		_V
30110	Total time to connect (seconds)		_V
30111	Inverter state (Same of 30074 register)		_W
30112	Cos(φ) **:** in thousandths (φ, delay angle between AC voltage and current) (Same as 30017)		_W
30113	Sign of sin(φ). 0 and 1 are for positive and negative values, respectively. (Same as 30018)		_W
30114	Power **:** Alternating output power generated by the INGECON-SUN, in **tens** of **Watt**. (Same as 30019)	yes	_W
30115	QAC (Reactive power DIV 10)	yes	_W

30116	1000V Kit usage Counter	-	_Y
30117	Power reduction register - Bit 0: Power limitation on (The inverter is not injecting all the available solar power). - Bit 1: Modbus power reduction on. - Bit 2: Max frequency power reduction on. - Bit 3: Grid fault power reduction on. - Bit 4: High Vdc power reduction on. - Bit 5: High temperature power reduction on.	-	_Z

2.1 Función 0x04: Input registers reading

Function 0x04 (Read Input Registers) is used to read online data from the inverter. The procedure starts with the PC sending a *Query* message to the inverter. The example below shows a message asking for data from 47 (0x2F) Input Registers. Note that 0x30001 range is a Modbus naming convention, and in practice register 0x30001 register is addressed at 0, and so on.

Address	--	Inverter Address[1 .. 247]
Function	0x04	Read Input Registers
Starting Address Hi	0x00	Address of 1st register (HI byte)
Starting Address Lo	0x00	Address of 1st register (LO byte)
Number of Points Hi	0x00	Number of registers to read (HI byte)
Number of Points Lo	0x2F	Number of registers to read (LO byte)
Error Check (CRC) - Hi	--	Cyclic redundancy code (HI byte)
Error Check (CRC) – Lo	--	Cyclic redundancy code (LO byte)

The inverter replies sending the following *Response* message, which in this case includes all possible variables (94 data bytes):

Address	0	--	Inverter Address[1 .. 247]
Function	1	0x04	Read Input Registers
Byte Count	2	0x5E	Number of data octets
Data 0	3	--	Value of *E_total* (byte 0, highest)
Data 1	4	--	Value of *E_total* (byte 1)
Data 2	5	--	Value of *E_total* (byte 2)
Data 3	6	--	Value of *E_total* (byte 3, lowest)
Data 0	7	--	Value of *Hours* (byte 0, highest)
Data 1	8	--	Value of *Hours* (byte 1)
Data 2	9	--	Value of *Hours* (byte 2)
Data 3	10	--	Value of *Hours* (byte 3, lowest)
Data 0	11	--	Value of *Conex* (byte 0, highest)
Data 1	12	--	Value of *Conex* (byte 1)
Data 2	13	--	Value of *Conex* (byte 2)
Data 3	14	--	Value of *Conex* (byte 3, lowest)
Data 0	15	--	Value of Alarmas_*Inv* (byte 0, highest)
Data 1	16	--	Value of Alarmas_*Inv* (byte 1)
Data 2	17	--	Value of Alarmas_*Inv* (byte 2)

Data 3	18	--	Value of Alarmas_*Inv* (byte 3, lowest)
Data Hi	19	--	Value of *VDC* (HI byte)
Data Lo	20	--	Value of *VDC* (LO byte)
Data Hi	21	--	Value of *IDC* (HI byte)
Data Lo	22	--	Value of *IDC* (LO byte)
Data Hi	23	--	Value of *VAC1* (HI byte)
Data Lo	24	--	Value of *VAC1* (LO byte)
Data Hi	25	--	Value of *VAC2*(HI byte)
Data Lo	26	--	Value of *VAC2* (LO byte)
Data Hi	27	--	Value of *VAC3*(HI byte)
Data Lo	28	--	Value of *VAC3* (LO byte)
Data Hi	29	--	Value of I*AC1* (HI byte)
Data Lo	30	--	Value of I*AC1* (LO byte)
Data Hi	31	--	Value of I*AC2*(HI byte)
Data Lo	32	--	Value of I*AC2* (LO byte)
Data Hi	33	--	Value of I*AC3*(HI byte)
Data Lo	34	--	Value of I*AC3* (LO byte)
Data Hi	35	--	Value of *CosFi* (HI byte)
Data Lo	36	--	Value of *CosFi* (LO byte)
Data Hi	37	--	Value of *SigSinFi* (HI byte)
Data Lo	38	--	Value of *SigSinFi* (LO byte)
Data Hi	39	--	Value of *PAC* (HI byte)
Data Lo	40	--	Value of *PAC* (LO byte)
Data Hi	41	--	Value of *FAC2* (HI byte)
Data Lo	42	--	Value of *FAC2* (LO byte)
Data Hi	43	--	Value of *F_year* (HI byte)
Data Lo	44	--	Value of *F_year* (LO byte)
Data Hi	45	--	Value of *F_Month* (HI byte)
Data Lo	46	--	Value of *F_Month* (LO byte)
Data Hi	47	--	Value of *F_Day* (HI byte)
Data Lo	48	--	Value of *F_Day* (LO byte)
Data Hi	49	--	Value of *H_Hour* (HI byte)
Data Lo	50	--	Value of *H_Hour* (LO byte)
Data Hi	51	--	Value of *H_Minutes* (HI byte)
Data Lo	52	--	Value of *H_Minutes* (LO byte)
Data Hi	53	--	Value of *H_Seconds* (HI byte)
Data Lo	54	--	Value of *H_Seconds* (LO byte)
Data Hi	55	--	Pos_grad_solPor10(HI byte)
Data Lo	56	--	Pos_grad_solPor10(LO byte)
Data Hi	57	--	Value of ModSeg_1 (HI byte)
Data Lo	58	--	Value of ModSeg_1 (LO byte)
Data Hi	59	--	Value of Est_MotSeg_1 (HI byte)
Data Lo	60	--	Value of Est_MotSeg_1 (LO byte)

Data Hi	61	--	Value of GPor10MotSeg_1 (HI byte)
Data Lo	62	--	Value of GPor10MotSeg_1 (LO byte)
Data Hi	63	--	Value of GPor10PosConsig_1 (HI byte)
Data Lo	64	--	Value of GPor10PosConsig_1 (LO byte)
Data Hi	65	--	Value of EstSeg_1 (HI byte)
Data Lo	66	--	Value of EstSeg_1 (LO byte)
Data Hi	67	--	Value of OffsetSeg_1 (HI byte)
Data Lo	68	--	Value of OffsetSeg_1 (LO byte)
Data Hi	69	--	Value of PasoSeg_1 (HI byte)
Data Lo	70	--	Value of PasoSeg_1 (LO byte)
Data Hi	71	--	Value of ModSeg_2 (HI byte)
Data Lo	72	--	Value of ModSeg_2 (LO byte)
Data Hi	73	--	Value of Est_MotSeg_2 (HI byte)
Data Lo	74	--	Value of Est_MotSeg_2 (LO byte)
Data Hi	75	--	Value of GPor10MotSeg_2 (HI byte)
Data Lo	76	--	Value of GPor10MotSeg_2 (LO byte)
Data Hi	77	--	Value of GPor10PosConsig_2 (HI byte)
Data Lo	78	--	Value of GPor10PosConsig_2 (LO byte)
Data Hi	79	--	Value of EstSeg_2 (HI byte)
Data Lo	80	--	Value of EstSeg_2 (LO byte)
Data Hi	81	--	Value of OffsetSeg_2 (HI byte)
Data Lo	82	--	Value of OffsetSeg_2 (LO byte)
Data Hi	83	--	Value of PasoSeg_2 (HI byte)
Data Lo	84	--	Value of PasoSeg_2 (LO byte)
Data Hi	85	--	Extra_Data_1 (HI byte)
Data Lo	86	--	Extra_Data_1 (LO byte)
Data Hi	87	--	Extra_Data_2 (HI byte)
Data Lo	88	--	Extra_Data_2 (LO byte)
Data Hi	89	--	Extra_Data_3 (HI byte)
Data Lo	90	--	Extra_Data_3 (LO byte)
Data Hi	91	--	Extra_Data_4 (HI byte)
Data Lo	92	--	Extra_Data_4 (LO byte)
Data Hi	93	--	Extra_Data_5 (HI byte)
Data Lo	94	--	Extra_Data_5 (LO byte)
Data Hi	95	--	Extra_Data_6 (HI byte)
Data Lo	96	--	Extra_Data_6 (LO byte)
Error Check (CRC) - Hi	97	--	Cyclic redundancy code (HI byte)
Error Check (CRC) - Lo	98	--	Cyclic redundancy code (LO byte)

2.2 Inverter alarm list

The alarms on the 30066 register are associated to the stop reasons on 30084 register. The interpretation of these alarms is explained in the table before.

Alarm		Stop reason	Description
0x0000		Ninguno	No alarms, the equipment should connect if it has enough power
0x0001	ALARMA_FRED	MOTIVO_PARO_FRED	Grid frequency out of range (49-51Hz)
0x0002	ALARMA_VRED	MOTIVO_PARO_VRED	Grid voltaje out of range (195V-253V)
0x0004	ALARMA_PI_ANA	MOTIVO_PARO_PI_ANA_SAT MOTIVO_PARO_PI_ANA_SAT	The current measure is much lower than the current checkpoint in that branch The initial PI in the integrator of the capture current does not work properly.
0x0008	ALARMA_RESET	MOTIVO_PARO_RESET_WD	Indicates that the invertir has been reset by Watch-Dog, failure in inverter's Firmware.
0x0010	ALARMA_IRED_EFICAZ	MOTIVO_PARO_IAC_EFICAZ	The current RMS value is greater than the máximum allowed.
0x0020	ALARMA_TEMPERATURA	MOTIVO_PARO_TEMPERATURA MOTIVO_PARO_TEMP_AUX	The power electronic temperatura is greater than 80 ºC. The temperature auxiliary sensor has detected an alarm.
0x0040	ALARMA_LEC_ADC	MOTIVO_PARO_ERROR_LEC_ADC MOTIVO_PARO_LATENCIA_ADC	Reading level in the ADC higher than normal in an input not expected. Intern error of the analogic digital converter
0x0080	ALARMA_IRED_INSTAN	MOTIVO_PARO_MAX_IAC_INST	Instantaneous current value out of range
0x0100	ALARMA_PROT_AC	MOTIVO_PARO_VARISTORES MOTIVO_PARO_CONTACTOR MOTIVO_PARO_PROT_AC MOTIVO_PARO_MAGNETO	AAS0043 AC varistor error The state of the contactor is not correct attending the inverter state. Error in the AC protections, dischargers, fuses… Error in the three phase input thermomagnetic (in big equipments)
0x0200	ALARMA_PROT_DC	MOTIVO_PARO_FUS_DC	DC input fuses melt or DC dischargers.
0x0400	ALARMA_AISL_DC	MOTIVO_PARO_AISL_DC	Isolation failure in the solar field or in the inverter interior
		MOTIVO_PARO_VARISTORES	DC varistors error
0x0800	ALARMA_FRAMA	MOTIVO_PARO_FRAMA1 MOTIVO_PARO_FRAMA2 MOTIVO_PARO_FRAMA3	Failure in the branch 1 of the power electronics Failure in the branch 2 of the power electronics Failure in the branch 3 of the power electronics
0x1000	ALARMA_PARO_MANUAL	MOTIVO_PARO_PARO_MANUAL	Manual Stop due to emergency push button, by display or communication
0x2000	ALARMA_CONFIG	MOTIVO_PARO_CONFIGURACION MOTIVO_PARO_CARGA_FIRMWARE	Stop due to Firmware modification. Stop due to Firmware load.
0x4000	ALARMA_VIN	MOTIVO_PARO_VIN	DC input input high voltage
0x8000	ALARMA_VPV_MED_MIN	MOTIVO_PARO_BAJA_VPV_MED	Stop because of low voltage in the input. The inverter controls this voltage, for this reason, this should never happen.

On the 30032 registers suntracker alarms are stored. If the firmware is a AAS1020 or AAS1040 these alarms have to be interpreted as follows:

Tracker alarms

0x01	Pulses	Too much time without pulses from encoder
0x02	Reference	Tracker lost reference
0x04	Limit switch	Limit switch pushed
0x08	Wind alarm	Too much wind detected
0x10	Pressure alarm	Oil pressure alarm

If the firmware is another firmware other signals are displayed in the same positions:

0x00xx	Reference power for the slaves. MS operation (in %) (only if MS inverter)
0x0080	Bit indicating that the 1000V Kit has worked.(only if 1000Vdc kit exists)
0x0100	Bit indicating that the inverter is the master of MS system. (only if MS inverter)
0x0200	Bit indicating that the inverter has working permission into the MS system. (only if MS inverter)
0x0400	Bit indicating that the inverter limits its power to avoid DC bus falling.
0x0800	Bit indicating that the inverter limits its power because of Modbus order.
0x1000	Bit indicating that the inverter limits its power because of high frequency.
0x2000	Bit indicating that the inverter limits its power because of grid voltage failure.
0x4000	Bit indicating that the inverter has detected a voltage dip.
0x8000	Bit indicating that the inverter has reduced Pac due to hi Vdc voltage.

3. SENDING COMMANDS (PRESET HOLDING REGISTERS)

Using the Modbus RTU protocol preset holding registers function commands also can be sent. On table 4 are represented command holding registers (between 41001 and 42000).

Modbus Register	Flash Ad (P)	Description	MIN	MAX	TYPE
41001		Command code	0	12	RW
41002		Command Data 1	-	-	RW
41003		Reserved	-	-	R
41004		Reserved	-	-	R
41005		Reserved	-	-	R
41006		Power reduction target	0	32767	R
41007		Phi tangent target	-15729	15729	R
41008		Reactive target	-32767	32767	R
41009		Reserved			R
41010		Restrictive Freq Limits	0	1	R
41011		Reserved			R
41012		Reserved			R

To send a command the command has to be written on 41001. The command code is represented on next table. Data has to be written on 41002 holding register if necessary. Writing in those two register using standard modbus (0x10 function) desired command will be applied.

After executing the command readable holding registers will be upgraded with new values.

NOTE: Starting address of command holding registers is 41001, on terms of sending frames it will be 1000 on dec or 0x03E8.

Command / Data table for Query:

Command number	**Command**	**Data parameter**	**Data parameter limits**
0	No command	(not used)	(not used)
1	Change phi tangent target	New phi tangent reference in Int16*	Max: 0.48 (15870) Min: -0.48 (-15870)
2	Read phi tangent target	(not used)	(not used)
3	Change power reduction target	Inverter Power in Uint16 **	Max: 100% (32767) Min: 0% (0)
4	Read power reduction target	(not used)	(not used)
5	Stop inverter	Sender Node number	Max: 254 Min: 1
6	Start Inverter	Sender Node number	Max: 254 Min: 1
7	No command	(not used)	(not used)
8	No command	(not used)	(not used)
9	Change reactive power ref	React. power in (KVAr/10)	Nominal power of the inverter div 10
10'	No command	(not used)	(not used)
11	Disable reactive power ref***	(not used)	(not used)
12	Enable restrictive frequency limits****	1: Restrictive limits ON 0: Restrictive limits OFF	Max: 1 Min: 0
13	Inject reactive power without DC source	React. power in (KVAr/10)	Nominal power of the inveter div 10
14	Stop reactive power injection without DC source	(not used)	(not used)

*****Int16 bit phi tangent: 32767 is equal to 1, -32767 I equal to -1. (Tan*32767)

******Uint16 Inverter power: 100% of max power: 32767. 0% of max power: 0 (%/100*32767) The maximum reduction is chosen, 0%, the inverter stops with “manual stop” alarm.

*** It changes to 0 the reactive power reference if previously is set to a given value

****Restrictive limits are 49,5Hz/50,5Hz. Non restrictive limits are 47,5Hz/51,5Hz. According to the CEI 0-21 regulation

Reactive Control works as a Generator. When the inverter is commanded with a positive tangent target, injected current will be delayed from voltage.

Otherwise, if tangent target is negative, injected current will be before the voltage.

If the INGECON SUN receives a command it resends the frame to the sender. On table 3 there are the reponse parameters

Command / Data table for Response:

Command number	**Command**	**Data parameter**	**Data parameter limits**
0	No command	(not used)	(not used)
1	Change phi tangent target	Current phi tangent reference in Int16	Max: 0.48 (15870) Min: -0.48 (-15870)
2	Read phi tangent target	Current phi tangent reference in Int16	Max: 0.48 (15870) Min: -0.48 (-15870)
3	Change power reduction target	Current Inverter Power in Uint16	Max: 100% (32767) Min: 5% (1638)
4	Read power reduction target	Current Inverter Power in Uint16	Max: 100% (32767) Min: 5% (1638)
5	Stop inverter	Stop-Start status	1: Stopped, 2:Started
6	Start Inverter	Stop-Start status	1: Stopped, 2:Started
7	No command	(not used)	(not used)
8	No command	(not used)	(not used)
9	Change reactive power ref	React. power in (KVAr/10)	Nominal power of the inveter div 10
10’	No command	(not used)	(not used)
11	Disable reactive power ref	(not used)	(not used)
12	Enable restrictive frequency limits	1: Restrictive limits ON 0: Restrictive limits OFF	(not used)
13	Inject reactive power without DC source	React. power in (KVAr/10)	Nominal power of the inveter div 10
14	Stop reactive power injection without DC source	(not used)	(not used)